

GTU CSE 241/501

FALL 2023

Midterm Exam

Student Code – Q1(20) – Q2(35) – Q3(45) – Exam Format(-10)
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Read the instructions below carefully.

- Do not write your name on your student number in any part of the exam materials including the answer sheets.
- Your individual student codes are already printed on your question and answer sheets. Keep this student code until the exam ends. You will use it to sign the exam attendance sheet.
- Answer each question on the designated page. Just put a single letter for each box. We will not (and cannot) grade any materials outside of the designated boxes.
- Hand over your cell phones, books, and notes to TA before the exam starts. You will not use any electronic devices or books during the exam.
- This exam has 3 questions; it will be graded out of 100 points. You have 120 minutes. If you do not follow any of the rules above, you will lose 10 points.

Question 1 (20)

- a) Explain why **assert** is implemented as macro instead of a global function?
- b) Why do we prefer the operator **new** over the function **malloc ()** in C++? Explain your answer.

Question 2 (35)

Point2D represents a two-dimensional point such as (3.5, 5.0). It has the following functions as well as any other functions that you think are necessary (setters, getters, etc.).

- A constructor that takes all parameters. All arguments should be defaulted.
- A function that returns the number of **Point2D** objects created so far.
- A function that returns the number of **Point2D** objects living at the time.
- Overloaded stream insertion operator (**cout << point**) will print **Point2D** details.
- Overloaded == and != operators for comparing **Point2D**'s.
- Overloaded pre and post increment operators that increment the point components by 1.0. For example, (1.2, 3.5) becomes (2.2, 4.5) after increment.
- Overloaded unary **operator!** That returns a new **Point2D** object by switching the X and Y component of the **Point2D**.
- Overloaded binary **operator-** that returns the Euclidian distance between the given points as a double value.

Q2 Part a) Write the header file **Point2D.h** for the above class. Do not implement any functions in the header other than very short one-line functions.

Q2 Part b) Write the implementation file **Point2D.cpp** for the above class.

Question 3 (45)

Study the following driver code for a class named **GTUSet**.

```
#include <iostream>
```

```
#include "GTUSet.h"
```

```
using std :: cout ;
```

```
int main ()    {  
    // Make set1 and insert numbers 1, 2, 3  
    GTU :: GTUSet set1 ;  
    set1.insert (1) ;  
    set1.insert (2) ;  
    set1.insert (3) ;  
    // Make set2 and insert numbers 3, 4, 5  
    GTU :: GTUSet set2 ;  
    set2.insert (3) ;  
    set2.insert (4) ;  
    set2.insert (5) ;  
    // Print the intersection of set1 and set2  
    // Prints 3  
    cout << (set1 ^ set2) << std :: endl ;  
    // Remove 1 from set1  
    set1.erase (1) ;  
    // Remove 5 from set2  
    set2.erase (5) ;  
    // Print the union of set1 and set2  
    // prints 2, 3, 4  
    cout << (set1 + set2) << std :: endl ;  
    // prints True  
    cout << (set1 + set2).find(3) << std :: endl ;  
    return 0 ;  
}
```

Class **GTUSet** has three data members only: an **int *** to keep the integers and two **ints** to keep the capacity and the size of the set.

Q3 Part a) Write the complete class header. Do not implement any functions in the header.

Q3 Part b) Write the implementation file that includes all necessary class member and global functions. Do not forget to check for errors. Remember, sets include only unique elements.